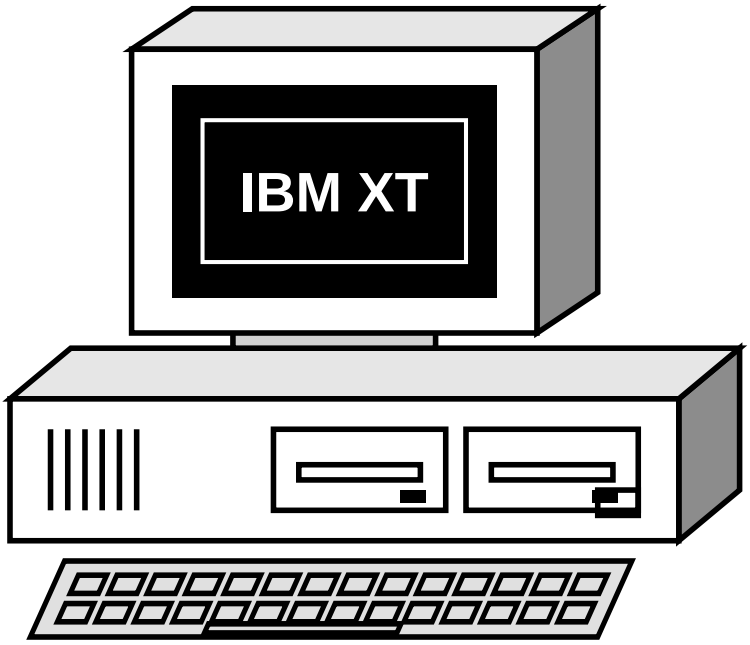
os8088

XT Guide

Installation and Operation



os8088 XT Guide - Installation and Operation

For the IBM Personal Computer XT and compatibles

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Conventions

Keys are written in capitals. Text you type is shown in Courier. A note in the margin marks something to check before you go on. Page numbers in the contents are links.

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COM1: 9600,N,8,1
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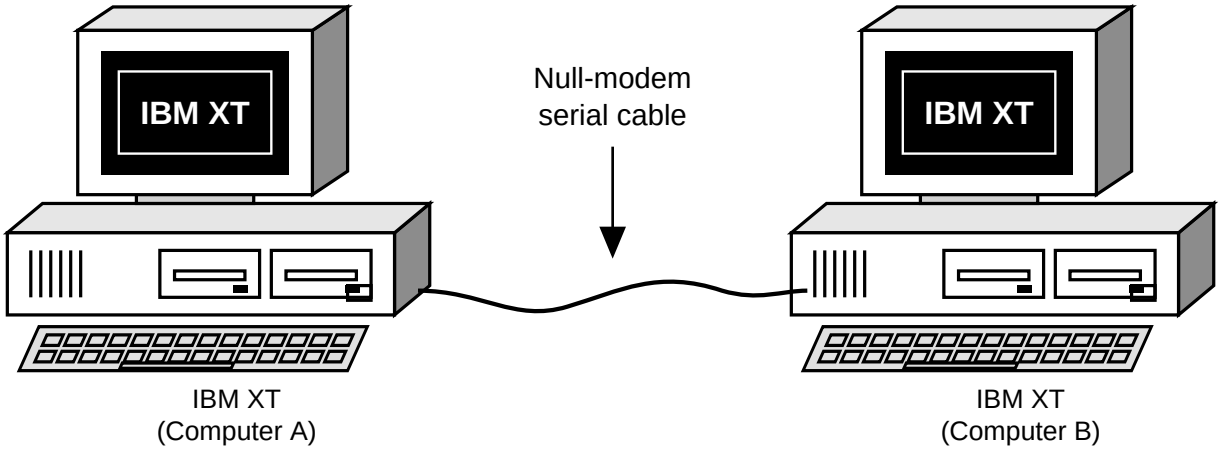
03 Connecting your XT

Link two computers.



Before you connect

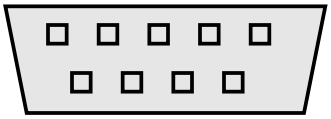
1. Shut down both computers.
2. Connect a serial cable.
3. Select the same baud rate.

Recommended settings

Port	COM1
Speed	9600 baud
Format	8 data bits, no parity, 1 stop bit

Serial pinout

DB-9 male connector on the serial card.



Pins 1-5 on top, 6-9 below

Pin	Signal	Name	Cable to pin
1	DCD	Carrier detect	-
2	RXD	Receive data	3
3	TXD	Transmit data	2
4	DTR	Terminal ready	6
5	GND	Signal ground	5
6	DSR	Data set ready	4
7	RTS	Request to send	8
8	CTS	Clear to send	7
9	RI	Ring indicator	-

A null-modem cable crosses transmit and receive, and the handshake pairs, so each computer listens where the other speaks. A straight cable will not work.

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04 Maintenance

Keep both machines in good order.

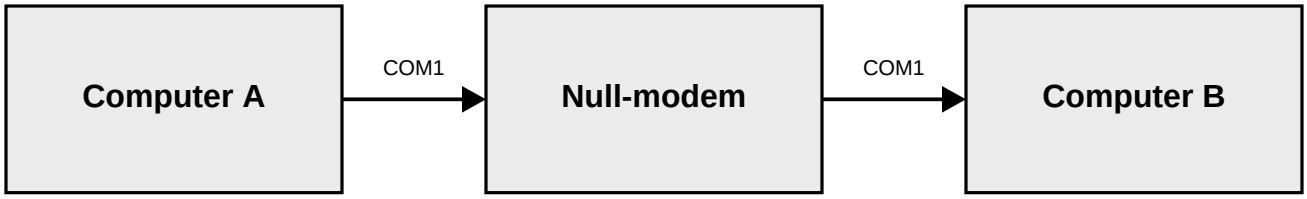
- ☐ Power both computers off before touching a cable.
- ☐ Clean the floppy heads every 40 hours of use.
- ☐ Check the card is seated and the screw is tight.
- ☐ Keep the cable away from the power supply fan.
- ☐ Vacuum the vents; do not blow dust into the drive.
- ☐ Label each disk and keep a spare boot copy.

Cleaning schedule

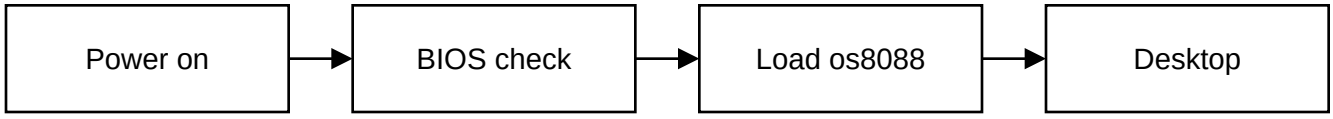
Part	How often	With
Floppy heads	Every 40 hours	Head cleaner disk
Keyboard	Monthly	Soft brush
Screen	Monthly	Dry cloth
Vents	Every 3 months	Vacuum

Diagrams

How the pieces connect.



Start-up sequence



05 Starting os8088

From power switch to desktop.

1. Insert the system disk in drive A.
2. Switch on the monitor first, then the computer.
3. Wait for the loading screen to finish.
4. Double-click a program to open it.
5. Choose Shut Down from the menu before switching off.

Booting from a floppy takes about a minute on a 4.77 MHz machine. The pointer appears as soon as the mouse driver is ready; do not press keys until the desktop is drawn.

Troubleshooting

Symptom	Likely cause	What to do
No picture	Monitor not on	Switch on, then reboot
Disk error	Dirty heads	Clean the heads
No link	Straight cable	Use a null-modem
Garbled text	Speeds differ	Match the baud rate
Mouse frozen	Wrong port	Move it to COM1

Keys and ports

Key	Action
ESC	Close the open menu
TAB	Move to the next field
F1	Open help
ALT+F4	Close the window
CTRL+ALT+DEL	Restart the machine

Ports on the back panel

Port	Address	Use
COM1	03F8	Serial mouse or link
COM2	02F8	Second serial device
LPT1	0378	Printer

Specifications

Item	Value
Processor	Intel 8088, 4.77 MHz
Memory	640 KiB
Display	VGA, CGA or Hercules
Storage	360 KB or 1.44 MB floppy
Serial	RS-232, 9600 baud

06 Glossary

Baud Symbols per second on a serial line.

COM1 The first serial port.

Null modem A cable that crosses transmit and receive.

Parity An error check bit; here, none.

Stop bit Marks the end of a character.

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